# Interactive Multi-Agent Requirement Refinement with Human-in-the-Loop

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## Abstract

Large language models can quickly draft software requirements, but the resulting requirements are often difficult for stakeholders to control, revise, and evolve. This paper studies human-in-the-loop multi-agent requirement refinement as an interactive requirements engineering problem. The proposed structure focuses on a workflow in which specialized agents elicit, analyze, and review requirements while human feedback and requirement knowledge guide iterative revision. The planned evaluation examines requirement quality, controllability, refinement effectiveness, user satisfaction, and human effort while keeping the study limited to requirement refinement.

## Keywords

large language models, multi-agent systems, human-in-the-loop, requirements engineering, requirement refinement, interactive requirement engineering

## 1 Introduction

### 1.1 Background and Motivation

Requirements are the main medium through which stakeholders express intent, constraints, and acceptable system behavior. Recent work has argued that requirements are becoming a central interface for intelligent software engineering [2], while requirement specification remains difficult because ambiguity, missing constraints, and domain vocabulary must still be handled in a stakeholder-readable form [1]. Large language models can draft requirements rapidly, but automatic drafting alone does not give users enough control over how requirements are revised.

### 1.2 Challenges of Interactive Requirement Refinement

This paper focuses on the refinement stage after an initial requirement draft is available. The key challenges are:

- **Limited controllability:** users often cannot precisely steer which parts of a requirement should be preserved, changed, constrained, or removed.
- **Weak feedback integration:** user comments are frequently treated as one-shot prompts instead of structured refinement signals.
- **Opaque evolution:** the rationale behind requirement revisions is often not visible to stakeholders.
- **Insufficient interaction:** requirement improvement requires clarification and negotiation, not only automatic generation.

### 1.3 Human-in-the-Loop Requirement Optimization

We study a human-in-the-loop multi-agent workflow for interactive requirement refinement. The workflow uses specialized agents to elicit clarification, analyze requirement quality, and review revisions, while human feedback remains the controlling signal for iterative optimization. Knowledge-assisted refinement is used to support revision templates, feedback patterns, and prompt strategies while keeping the scope limited to requirement refinement.

### 1.4 Contributions

The paper is organized around the following contributions:

- We formulate human-in-the-loop requirement refinement as a controllability problem in intelligent requirements engineering.
- We design a lightweight multi-agent interactive workflow for requirement revision, constraint adjustment, and feedback integration.
- We introduce knowledge-assisted refinement mechanisms that guide iterative requirement optimization.
- We define an evaluation plan centered on requirement quality, controllability, refinement effectiveness, user satisfaction, and human effort.

## 2 Related Work

### 2.1 Human-in-the-Loop Systems

Human-in-the-loop systems emphasize continuous user oversight, intervention, and correction. For requirement refinement, the relevant issue is not only whether a human can approve an output, but whether the user can guide revisions during an evolving requirement process.

### 2.2 Interactive Requirement Engineering

Interactive requirement engineering studies how stakeholders clarify goals, negotiate constraints, and iteratively improve requirement statements. This work builds on that perspective by treating interaction as a core refinement mechanism rather than an optional interface layer.

### 2.3 Requirement Refinement

Requirement refinement concerns transforming vague, incomplete, or inconsistent requirement drafts into clearer and more actionable specifications. The focus of this paper is limited to requirement-level revision, constraint adjustment, and evolution under user feedback.

### 2.4 LLM-Based Requirement Engineering

LLM-based requirement engineering explores how generative models can support elicitation, drafting, rewriting, and analysis of requirements. Existing automatic approaches are useful for producing candidate requirements, but they often provide limited support for controlled, transparent, and iterative refinement.

## 2.5 Multi-Agent and Human-AI Collaboration

Multi-agent collaboration can distribute responsibilities such as elicitation, analysis, and review. Human-AI collaboration further requires that users remain able to inspect, correct, and redirect the refinement process. This paper combines these two lines of work in a lightweight workflow for interactive requirement optimization.

## 2.6 Positioning of This Work

Unlike work that treats requirements as input to later development tasks, this paper focuses only on requirement refinement itself. The contribution is a research-problem-driven structure for studying how human feedback, agent collaboration, and knowledge assistance improve requirement controllability.

# 3 Methodology

## 3.1 Framework Overview

The framework refines a draft requirement through repeated user-agent interaction and a lightweight requirement knowledge base. It takes a user goal or draft as input and produces an evolved requirement version with feedback, rationale, and open issues.

## 3.2 Multi-Agent Interactive Workflow

The workflow is intentionally small and task-oriented. Each agent has a bounded role so that the human user can understand and control the refinement process.

*3.2.1 Interviewer Agent.* The Interviewer Agent asks clarification questions when user intent, domain context, priority, or constraints are underspecified.

*3.2.2 Analyst Agent.* The Analyst Agent identifies ambiguity, missing conditions, vague quality attributes, and refinement opportunities in the current requirement version.

*3.2.3 Reviewer Agent.* The Reviewer Agent checks whether a proposed revision follows the user's feedback and whether it preserves the intended requirement meaning.

*3.2.4 Human Feedback Loop.* The human user can accept, reject, edit, or redirect revisions. Feedback is stored as a refinement signal and used in later iterations.

## 3.3 Requirement Refinement Process

The refinement process is modeled as an iterative cycle rather than a one-shot generation step.

*3.3.1 Requirement Revision.* Requirement text is revised to improve clarity, completeness, testability, and stakeholder readability while keeping the user's intent explicit.

*3.3.2 Constraint Adjustment.* Functional constraints, quality constraints, scope boundaries, and priorities can be added, relaxed, or narrowed according to user feedback.

*3.3.3 User Feedback Integration.* User comments are normalized into refinement actions such as clarify, specialize, generalize, preserve, remove, or reprioritize.

*3.3.4 Requirement Evolution.* Each iteration records the requirement version, user feedback, agent suggestion, accepted revision, and rationale for change.

## 3.4 Knowledge-Assisted Requirement Optimization

Knowledge assistance supports refinement without replacing user judgment. It provides reusable guidance for common requirement improvement patterns.

*3.4.1 Requirement Knowledge.* Requirement knowledge includes domain terminology, requirement quality criteria, common ambiguity patterns, and stakeholder constraints.

*3.4.2 Refinement Templates.* Refinement templates guide recurring edits such as adding actors, conditions, acceptance criteria, priority, and measurable quality attributes.

*3.4.3 Feedback Patterns.* Feedback patterns map user instructions to concrete refinement operations, making the revision process more transparent and repeatable.

*3.4.4 Prompt Strategies.* Prompt strategies constrain agent behavior to ask targeted questions, preserve user intent, and explain revision rationale.

## 3.5 Human-in-the-Loop Interaction

Human participation is treated as a first-class part of the method.

*3.5.1 User Intervention.* Users can intervene when an agent misunderstands intent, over-refines a requirement, ignores constraints, or changes requirement scope.

*3.5.2 Interactive Clarification.* The system asks clarification questions only when the missing information affects refinement decisions.

*3.5.3 Iterative Feedback.* Feedback from earlier iterations is reused to keep later revisions consistent with stakeholder preferences.

## 3.6 Requirement Controllability

Requirement controllability refers to the user's ability to steer what changes, why it changes, and how the revised requirement evolves over iterations. We operationalize controllability through edit control, constraint control, rationale visibility, version comparison, and feedback traceability.

# 4 Experiments

## 4.1 Research Questions

The evaluation is organized around the effectiveness of the interactive requirement refinement workflow:

- **RQ1:** Does human-in-the-loop multi-agent refinement improve requirement quality compared with non-interactive requirement drafting?
- **RQ2:** Does structured user feedback improve requirement controllability during iterative refinement?
- **RQ3:** How much do knowledge-assisted refinement mechanisms contribute to refinement effectiveness?

- **RQ4:** How do users perceive the interaction cost, transparency, and usefulness of the workflow?

### 4.2 Experimental Setup

This subsection will describe the study protocol, participant roles, task procedure, interaction budget, model configuration, and refinement iteration limit. The setup should keep the evaluation focused on requirement refinement rather than later software engineering activities.

### 4.3 Datasets

The dataset should contain requirement drafts or stakeholder scenarios that require clarification, constraint adjustment, and iterative revision. Each item should include the initial requirement context, user feedback opportunities, and reference criteria for evaluation.

### 4.4 Baselines

Baselines should represent reasonable alternatives for requirement refinement:

- **Single-pass LLM:** produces a refined requirement without interaction.
- **Single-agent interactive refinement:** uses user feedback but no role-based agent collaboration.
- **Multi-agent without knowledge assistance:** uses the agent workflow without refinement templates or feedback patterns.

### 4.5 Evaluation Metrics

The evaluation will measure both artifact quality and interaction quality:

- **Requirement Quality:** clarity, completeness, consistency, testability, and readability.
- **User Satisfaction:** perceived usefulness, transparency, and confidence in the refined requirement.
- **Requirement Controllability:** ability to preserve intent, direct edits, manage constraints, and inspect revision rationale.
- **Refinement Effectiveness:** improvement between initial and final requirement versions.
- **Human Effort Reduction:** number of interaction turns, editing time, and perceived workload.

### 4.6 Main Results

This subsection will report quantitative and qualitative findings after the experiment is conducted. No result values are assumed in the current paper skeleton.

### 4.7 Ablation Study

The ablation study will isolate the contribution of the Human Feedback Loop, Reviewer Agent, and knowledge-assisted refinement components.

### 4.8 Case Study

The case study will trace one requirement refinement session from initial draft to final revision, emphasizing user interventions, agent suggestions, accepted changes, rejected changes, and revision rationale.

## 5 Threats to Validity

### 5.1 Internal Validity

Internal validity threats may arise from differences in participant experience, task familiarity, feedback quality, and model response variability. The study should mitigate these threats through consistent task instructions, controlled interaction budgets, and repeated evaluation where feasible.

### 5.2 External Validity

External validity threats concern whether the findings generalize beyond the selected requirement scenarios, domains, participants, and model settings. The evaluation should therefore include requirements with different ambiguity types, constraint types, and stakeholder goals.

### 5.3 Construct Validity

Construct validity threats arise if metrics such as controllability, satisfaction, or refinement effectiveness do not accurately capture the intended concepts. The study should combine expert assessment, user questionnaires, revision trace analysis, and clearly defined scoring rubrics.

## 6 Conclusion

This paper frames human-in-the-loop multi-agent requirement refinement as a focused research problem in intelligent requirements engineering. The proposed structure studies how users, specialized agents, and requirement knowledge can jointly support interactive, iterative, and controllable requirement optimization. The planned evaluation centers on requirement quality, controllability, refinement effectiveness, user satisfaction, and human effort, keeping the scope limited to requirement refinement.

## References

[1] Ikram Darif, Ghizlane El Boussaidi, Segla Kpodjedo, and Cristiano Politowski. 2026. Controlled Natural Language for Requirements Specification: A Systematic Literature Review. *ACM Comput. Surv.* 58, 7 (Jan. 2026), 175:1–175:36. doi:10.1145/3778169

[2] Diana Robinson, Christian Cabrera, Andrew D. Gordon, Neil D. Lawrence, and Lars Mennen. 2025. Requirements Are All You Need: The Final Frontier for End-User Software Engineering. *ACM Trans. Softw. Eng. Methodol.* 34, 5 (May 2025), 141:1–141:22. doi:10.1145/3708524

## A Appendix

The appendix may include supplementary materials for the study protocol, questionnaire items, requirement quality rubric, refinement trace examples, and prompt templates used by the agents.